

Internal Dependencies

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List_all_Java_artifacts

Table 1a - Top 30 artifacts with the highest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-4.12.1.jar	70	830	8	0
1	axon-server-connector-4.12.1.jar	11	147	1	4
2	axon-modelling-4.12.1.jar	10	158	6	1
3	axon-eventsourcing-4.12.1.jar	9	133	5	2
4	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
5	axon-test-4.12.1.jar	8	87	1	3
6	axon-configuration-4.12.1.jar	1	43	2	4
7	axon-disruptor-4.12.1.jar	1	22	1	3
8	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1

Table 1b - Top 30 artifacts with the highest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-4.12.1.jar	70	830	8	0
1	axon-modelling-4.12.1.jar	10	158	6	1
2	axon-server-connector-4.12.1.jar	11	147	1	4
3	axon-eventsourcing-4.12.1.jar	9	133	5	2
4	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
5	axon-test-4.12.1.jar	8	87	1	3
6	axon-configuration-4.12.1.jar	1	43	2	4
7	axon-disruptor-4.12.1.jar	1	22	1	3
8	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-4.12.1.jar	70	830	8	0
1	axon-modelling-4.12.1.jar	10	158	6	1
2	axon-eventsourcing-4.12.1.jar	9	133	5	2
3	axon-configuration-4.12.1.jar	1	43	2	4
4	axon-disruptor-4.12.1.jar	1	22	1	3
5	axon-server-connector-4.12.1.jar	11	147	1	4
6	axon-test-4.12.1.jar	8	87	1	3
7	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
8	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7

Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
1	axon-configuration-4.12.1.jar	1	43	2	4
2	axon-server-connector-4.12.1.jar	11	147	1	4
3	axon-disruptor-4.12.1.jar	1	22	1	3
4	axon-test-4.12.1.jar	8	87	1	3
5	axon-eventsourcing-4.12.1.jar	9	133	5	2
6	axon-modelling-4.12.1.jar	10	158	6	1
7	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
8	axon-messaging-4.12.1.jar	70	830	8	0

Table 1e - Top 30 artifacts with the lowest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-configuration-4.12.1.jar	1	43	2	4
1	axon-disruptor-4.12.1.jar	1	22	1	3
2	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
3	axon-test-4.12.1.jar	8	87	1	3
4	axon-eventsourcing-4.12.1.jar	9	133	5	2
5	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
6	axon-modelling-4.12.1.jar	10	158	6	1
7	axon-server-connector-4.12.1.jar	11	147	1	4
8	axon-messaging-4.12.1.jar	70	830	8	0

Table 1f - Top 30 artifacts with the lowest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
1	axon-disruptor-4.12.1.jar	1	22	1	3
2	axon-configuration-4.12.1.jar	1	43	2	4
3	axon-test-4.12.1.jar	8	87	1	3
4	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
5	axon-eventsourcing-4.12.1.jar	9	133	5	2
6	axon-server-connector-4.12.1.jar	11	147	1	4
7	axon-modelling-4.12.1.jar	10	158	6	1
8	axon-messaging-4.12.1.jar	70	830	8	0

Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7
1	axon-disruptor-4.12.1.jar	1	22	1	3
2	axon-server-connector-4.12.1.jar	11	147	1	4
3	axon-test-4.12.1.jar	8	87	1	3
4	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
5	axon-configuration-4.12.1.jar	1	43	2	4
6	axon-eventsourcing-4.12.1.jar	9	133	5	2
7	axon-modelling-4.12.1.jar	10	158	6	1
8	axon-messaging-4.12.1.jar	70	830	8	0

Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-4.12.1.jar	70	830	8	0
1	axon-modelling-4.12.1.jar	10	158	6	1
2	axon-tracing-opentelemetry-4.12.1.jar	1	5	1	1
3	axon-eventsourcing-4.12.1.jar	9	133	5	2
4	axon-disruptor-4.12.1.jar	1	22	1	3
5	axon-test-4.12.1.jar	8	87	1	3
6	axon-configuration-4.12.1.jar	1	43	2	4
7	axon-server-connector-4.12.1.jar	11	147	1	4
8	axon-spring-boot-autoconfigure-4.12.1.jar	9	90	0	7

Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies](#)

Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBackward
0	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	0.900000	19	1
1	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	0.882353	16	1
2	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.tracing	0.875000	15	1
3	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging	0.857143	39	3
4	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging.annotation	0.840000	23	2
5	axon-messaging-4.12.1	org.axonframework.deadline	axon-messaging-4.12.1	org.axonframework.tracing	0.800000	9	1
6	axon-messaging-4.12.1	org.axonframework.commandhandling	axon-messaging-4.12.1	org.axonframework.tracing	0.777778	8	1
7	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing.eventstore	0.777778	16	2
8	axon-messaging-4.12.1	org.axonframework.commandhandling.callbacks	axon-messaging-4.12.1	org.axonframework.commandhandling	0.733333	13	2
9	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.serialization	0.647059	14	3
10	axon-messaging-4.12.1	org.axonframework.messaging.unitofwork	axon-messaging-4.12.1	org.axonframework.messaging	0.647059	14	3
11	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	0.615385	21	5
12	axon-messaging-4.12.1	org.axonframework.eventhandling.async	axon-messaging-4.12.1	org.axonframework.eventhandling	0.571429	11	3
13	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	0.555556	21	6
14	axon-messaging-4.12.1	org.axonframework.eventhandling.replay	axon-messaging-4.12.1	org.axonframework.eventhandling	0.454545	8	3
15	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.util	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	0.428571	5	2
16	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.util	0.333333	2	1
17	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query.subscription	0.333333	4	2
18	axon-messaging-4.12.1	org.axonframework.serialization.upcasting.event	axon-messaging-4.12.1	org.axonframework.eventhandling	0.333333	6	3
19	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing.eventstore.jdbc	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing.eventstore.jdbc.statements	0.317073	27	14
20	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.eventhandling.tokenstore	0.285714	9	5
21	axon-modelling-	org.axonframework.modelling.command.inspection	axon-modelling-4.12.1	org.axonframework.modelling.command	0.250000	20	12

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBackward
	4.12.1						
22	axon-messaging-4.12.1	org.axonframework.queryhandling.registration	axon-messaging-4.12.1	org.axonframework.queryhandling	0.250000	5	3
23	axon-messaging-4.12.1	org.axonframework.messaging	axon-messaging-4.12.1	org.axonframework.serialization	0.238095	13	8
24	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.command	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	0.222222	11	7
25	axon-messaging-4.12.1	org.axonframework.messaging.annotation	axon-messaging-4.12.1	org.axonframework.messaging.interceptors	0.200000	3	2
26	axon-modelling-4.12.1	org.axonframework.modelling.saga	axon-modelling-4.12.1	org.axonframework.modelling.saga.metamodel	0.142857	4	3
27	axon-messaging-4.12.1	org.axonframework.commandhandling.distributed.commandfilter	axon-messaging-4.12.1	org.axonframework.commandhandling.distributed	0.076923	7	6

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies_Breakdown](#)

Columns in addition to Table 2a:

- dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency	forwardToBackwardBalance	numberForward	numberBackward
0	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventProcessor->Span	0.900000	19	1
1	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventProcessorSpanFactory->Span	0.900000	19	1
2	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	EventProcessorSpanFactory->Span	0.900000	19	1
3	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	SimpleEventBus\$Builder->SpanFactory	0.900000	19	1
4	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	EventMessage<-NestingSpanFactory	0.900000	19	1
5	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	SubscribingEventProcessors\$Builder->SpanFactory	0.900000	19	1
6	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventBusSpanFactory->SpanFactory	0.900000	19	1
7	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventProcessorSpanFactory\$Builder->SpanFactory	0.900000	19	1
8	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventProcessorSpanFactory->SpanFactory	0.900000	19	1
9	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventBus->SpanScope	0.900000	19	1
10	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	EventBusSpanFactory->Span	0.900000	19	1
11	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventBusSpanFactory\$Builder->SpanFactory	0.900000	19	1
12	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventBus\$Builder->NoOpSpanFactory	0.900000	19	1
13	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventProcessors\$Builder->NoOpSpanFactory	0.900000	19	1
14	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventBus\$Builder->SpanFactory	0.900000	19	1
15	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventBusSpanFactory->Span	0.900000	19	1
16	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventBus->Span	0.900000	19	1
17	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	AbstractEventProcessors\$Builder->SpanFactory	0.900000	19	1
18	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	TrackingEventProcessors\$Builder->SpanFactory	0.900000	19	1
19	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultEventProcessorSpanFactory->NoOpSpanFactory\$NoOpSpan	0.900000	19	1
20	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	GenericStreamingQueryMessage->PublisherResponseType	0.882353	16	1
21	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	StreamingQueryMessage->ResponseType	0.882353	16	1
22	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	GenericQueryMessage->ResponseType	0.882353	16	1
23	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SimpleQueryUpdateEmitter->ResponseType	0.882353	16	1
24	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SimpleQueryUpdateEmitter->OptionalResponseType	0.882353	16	1
25	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SimpleQueryUpdateEmitter->MultipleInstancesResponseType	0.882353	16	1
26	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SimpleQueryBus->ResponseType	0.882353	16	1
27	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	GenericSubscriptionQueryMessage->ResponseType	0.882353	16	1
28	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QueryResponseMessage<-ConvertingResponseMessage	0.882353	16	1
29	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	GenericStreamingQueryMessage->ResponseType	0.882353	16	1
30	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QuerySubscription->ResponseType	0.882353	16	1
31	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QueryMessage->ResponseType	0.882353	16	1
32	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SimpleQueryUpdateEmitter->PublisherResponseType	0.882353	16	1
33	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QueryGateway->ResponseTypes	0.882353	16	1
34	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QueryGateway->ResponseType	0.882353	16	1
35	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	SubscriptionQueryMessage->ResponseType	0.882353	16	1
36	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	DefaultQueryGateway->ResponseType	0.882353	16	1
37	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.tracing	DefaultQueryBusSpanFactory\$Builder->SpanFactory	0.875000	15	1
38	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.tracing	SimpleQueryUpdateEmitter->Span	0.875000	15	1
39	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.tracing	SimpleQueryBus->SpanScope	0.875000	15	1

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown_BackwardOnly

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency	forwardToBackward
0	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.tracing	EventMessage<- NestingSpanFactory	
1	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.messaging.responsetypes	QueryResponseMessage<- ConvertingResponseMessage	
2	axon-messaging-4.12.1	org.axonframework.queryhandling	axon-messaging-4.12.1	org.axonframework.tracing	QueryMessage<- SpanUtils	
3	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging	DomainEventMessage<- Headers	
4	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging	TrackingToken<- StreamableMessageSource	
5	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging	EventMessage<- Headers	
6	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging.annotation	DomainEventMessage<- SourceIdParameterResolverFactory\$SourceIdParameterResolver	
7	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.messaging.annotation	AggregateTypeParameterResolverFactory\$AggregateTypeParameterResolver	
8	axon-messaging-4.12.1	org.axonframework.deadline	axon-messaging-4.12.1	org.axonframework.tracing	DeadlineMessage<- SpanUtils	
9	axon-messaging-4.12.1	org.axonframework.commandhandling	axon-messaging-4.12.1	org.axonframework.tracing	CommandMessage<- SpanUtils	
10	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing.eventstore	EventStreamUtils<- AbstractEventStorageEngine	
11	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing	axon-eventsourcing-4.12.1	org.axonframework.eventsourcing.eventstore	EventStreamUtils<- DomainEventStream	
12	axon-messaging-4.12.1	org.axonframework.commandhandling.callbacks	axon-messaging-4.12.1	org.axonframework.commandhandling	LoggingCallback<- SimpleCommandBus\$Builder	
13	axon-messaging-4.12.1	org.axonframework.commandhandling.callbacks	axon-messaging-4.12.1	org.axonframework.commandhandling	NoOpCallback<- SimpleCommandBus\$Builder	
14	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.serialization	GenericEventMessage<- AbstractXStreamSerializer	
15	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.serialization	GenericDomainEventMessage<- AbstractXStreamSerializer	
16	axon-messaging-4.12.1	org.axonframework.eventhandling	axon-messaging-4.12.1	org.axonframework.serialization	GapAwareTrackingToken<- GapAwareTrackingTokenConverter	
17	axon-messaging-4.12.1	org.axonframework.messaging.unitofwork	axon-messaging-4.12.1	org.axonframework.messaging	UnitOfWork<- DefaultInterceptorChain	
18	axon-messaging-4.12.1	org.axonframework.messaging.unitofwork	axon-messaging-4.12.1	org.axonframework.messaging	CurrentUnitOfWork<- GenericMessage	
19	axon-messaging-4.12.1	org.axonframework.messaging.unitofwork	axon-messaging-4.12.1	org.axonframework.messaging	UnitOfWork<- MessageHandlerInterceptor	
20	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerEventStoreFactory\$Builder<- ServerConnectorConfigurerModule	
21	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerEventStore<- ServerConnectorConfigurerModule	
22	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerEventStoreFactory<- ServerConnectorConfigurerModule	
23	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerEventStore\$Builder<- ServerConnectorConfigurerModule	
24	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.axon	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	EventProcessorInfoConfiguration<- ServerConnectorConfigurerModule	
25	axon-messaging-4.12.1	org.axonframework.eventhandling.async	axon-messaging-4.12.1	org.axonframework.eventhandling	SequencingPolicy<- SimpleEventHandlerInvoker\$Builder	
26	axon-messaging-4.12.1	org.axonframework.eventhandling.async	axon-messaging-4.12.1	org.axonframework.eventhandling	SequentialPerAggregatePolicy<- SimpleEventHandlerInvoker\$Builder	
27	axon-messaging-4.12.1	org.axonframework.eventhandling.async	axon-messaging-4.12.1	org.axonframework.eventhandling	SequencingPolicy<- SimpleEventHandlerInvoker	
28	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerQueryBus<- ServerConnectorConfigurerModule	
29	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerQueryBus\$Builder<- ServerConnectorConfigurerModule	
30	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerNonTransientRemoteQueryHandlingException<- ErrorCode	
31	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerRemoteQueryHandlingException<- ErrorCode	
32	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	AxonServerQueryDispatchException<- ErrorCode	
33	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.query	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	QueryPriorityCalculator<- ServerConnectorConfigurerModule	
34	axon-messaging-4.12.1	org.axonframework.eventhandling.replay	axon-messaging-4.12.1	org.axonframework.eventhandling	ResetContext<- ResetHandler	
35	axon-messaging-4.12.1	org.axonframework.eventhandling.replay	axon-messaging-4.12.1	org.axonframework.eventhandling	GenericResetContext<- AnnotationEventHandlerAdapter	
36	axon-messaging-4.12.1	org.axonframework.eventhandling.replay	axon-messaging-4.12.1	org.axonframework.eventhandling	ResetContext<- AnnotationEventHandlerAdapter	
37	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.util	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	ExceptionSerializer<- ErrorCode	
38	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.util	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	GrpcMessageSizeInterceptor<- AxonServerConnectionManager\$Builder	
39	axon-server-connector-4.12.1	org.axonframework.axonserver.connector	axon-server-connector-4.12.1	org.axonframework.axonserver.connector.event.util	ErrorCode<- GrpcExceptionParser	

Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

	fullDependentTypeName	declaredMethods	calledMethodNames	calledMethods	callerTypes
0	org.axonframework.commandhandling.CommandMessage	9	[getCommandName]	1	20
1	org.axonframework.eventhandling.EventMessage	9	[getIdentifier, getTimestamp]	2	11
2	org.axonframework.eventhandling.EventMessage	9	[getIdentifier]	1	10
3	org.axonframework.eventhandling.DomainEventMessage	10	[getSequenceNumber]	1	9
4	org.axonframework.eventhandling.TrackedEventMessage	10	[trackingToken]	1	8
5	org.axonframework.commandhandling.GenericCommandResultMessage	14	[asCommandResultMessage]	1	6
6	org.axonframework.eventhandling.DomainEventMessage	10	[getSequenceNumber, getType, getAggregateIdentifier]	3	6
7	org.axonframework.messaging.ResultMessage	9	[exceptionResult, isExceptional]	2	6
8	org.axonframework.eventhandling.ReplayToken	13	[createReplayToken]	1	5
9	org.axonframework.deadline.GenericDeadlineMessage	11	[asDeadlineMessage]	1	5
10	org.axonframework.eventhandling.DomainEventMessage	10	[getAggregateIdentifier]	1	5
11	org.axonframework.eventhandling.TrackedEventMessage	12	[trackingToken]	1	4
12	org.axonframework.eventhandling.GenericEventMessage	11	[asEventMessage]	1	4
13	org.axonframework.deadline.DeadlineMessage	10	[getDeadlineName]	1	4
14	org.axonframework.eventhandling.DomainEventMessage	10	[getType]	1	4
15	org.axonframework.eventhandling.EventMessage	9	[getTimestamp]	1	4
16	org.axonframework.deadline.DefaultDeadlineManagerSpanFactory	8	[builder]	1	4
17	org.axonframework.common.transaction.NoTransactionManager	4	[instance]	1	4
18	org.axonframework.queryhandling.SimpleQueryUpdateEmitter	17	[builder]	1	3
19	org.axonframework.commandhandling.GenericCommandResultMessage	15	[asCommandResultMessage]	1	3
20	org.axonframework.messaging.annotation.WrappedMessageHandlingMember	14	[handle]	1	3
21	org.axonframework.modelling.command.inspection.AggregateModel	13	[type]	1	3
22	org.axonframework.queryhandling.SubscriptionQueryMessage	12	[getUpdateResponseType]	1	3
23	org.axonframework.eventhandling.DomainEventMessage	11	[getSequenceNumber, getType, getAggregateIdentifier]	3	3
24	org.axonframework.queryhandling.SimpleQueryBus	11	[builder]	1	3
25	org.axonframework.eventhandling.DomainEventMessage	10	[getSequenceNumber, getAggregateIdentifier]	2	3
26	org.axonframework.eventhandling.GapAwareTrackingToken	10	[newInstance, withGapsTruncatedAt, advanceTo, getIndex, getGaps]	5	3
27	org.axonframework.eventhandling.GenericEventMessage	10	[asEventMessage]	1	3
28	org.axonframework.queryhandling.DefaultQueryBusSpanFactory	10	[builder]	1	3
29	org.axonframework.commandhandling.gateway.DefaultCommandGateway	9	[builder]	1	3
30	org.axonframework.config.Configuration	9	[getComponent]	1	3
31	org.axonframework.config.Configuration	9	[upcasterChain, snapshotFilter]	2	3
32	org.axonframework.messaging.MessageDecorator	9	[describeTo]	1	3
33	org.axonframework.commandhandling.DefaultCommandBusSpanFactory	5	[builder]	1	3
34	org.axonframework.eventhandling.TrackedEventData	5	[trackingToken]	1	3
35	org.axonframework.eventhandling.tokenstore.ConfigToken	5	[get]	1	3
36	org.axonframework.eventhandling.DefaultEventBusSpanFactory	4	[builder]	1	3
37	org.axonframework.queryhandling.DefaultQueryUpdateEmitterSpanFactory	4	[builder]	1	3
38	org.axonframework.modelling.command.inspection.AnnotatedAggregate	17	[initSequence, initialize, getAggregateRoot]	4	2
39	org.axonframework.queryhandling.DefaultQueryGateway	17	[builder]	1	2

Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
0	org.axonframework.common.BuilderUtils	BuilderUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Ma...	52
1	org.axonframework.common.AxonConfigurationException	AxonConfigurationException	[Type, File, Java, Class, ByteCode, Throwable, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedCom...	43
2	org.axonframework.messaging.Message	Message	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInduce...	42
3	org.axonframework.messaging.MetaData	MetaData	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4Ty...	40
4	org.axonframework.serialization.Serializer	Serializer	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	37
5	org.axonframework.eventhandling.EventMessage	EventMessage	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeakl...	36
6	org.axonframework.messaging.unitofwork.UnitOfWork	UnitOfWork	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWe...	33
7	org.axonframework.common.transaction.TransactionManager	TransactionManager	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2...	31
8	org.axonframework.common.Assert	Assert	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Ma...	29
9	org.axonframework.serialization.SerializedObject	SerializedObject	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInduce...	28
10	org.axonframework.serialization.SerializedType	SerializedType	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0]...	28
11	org.axonframework.messaging.unitofwork.CurrentUnitOfWork	CurrentUnitOfWork	[Type, File, Java, Class, ByteCode, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation36, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut36, Mark4TypeMaximumKCut...	22
12	org.axonframework.common.Registration	Registration	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6, M...	22
13	org.axonframework.tracing.SpanFactory	SpanFactory	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, ...	22
14	org.axonframework.lifecycle.Lifecycle	Lifecycle	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity0, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut36, Mark4TypeHDBSCAN97]	21
15	org.axonframework.lifecycle.Lifecycle\$LifecycleRegistry	Lifecycle\$LifecycleRegistry	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity0, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut59, Mark4TypeHDBSCAN1]	21
16	org.axonframework.messaging.MessageHandlerInterceptor	MessageHandlerInterceptor	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity3, Mark4Ty...	20
17	org.axonframework.common.ObjectUtils	ObjectUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut86, Mark4TypeHDBSCAN1]	20
18	org.axonframework.eventhandling.TrackingToken	TrackingToken	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0]...	20
19	org.axonframework.eventhandling.DomainEventMessage	DomainEventMessage	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation32, Mark4T...	19
20	org.axonframework.messaging.MessageDispatchInterceptor	MessageDispatchInterceptor	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation15, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity10, Mark4Type...	19
21	org.axonframework.messaging.annotation.ParameterResolverFactory	ParameterResolverFactory	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, ...	19
22	org.axonframework.serialization.SimpleSerializedObject	SimpleSerializedObject	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation16, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity12, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut96, Mark4Ty...	19
23	org.axonframework.common.AxonNonTransientException	AxonNonTransientException	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation10, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecompositi...	18
24	org.axonframework.eventhandling.GenericEventMessage	GenericEventMessage	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation18, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity12, Mark4TypeKCor...	18
25	org.axonframework.commandhandling.CommandMessage	CommandMessage	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeakl...	17
26	org.axonframework.tracing.NoOpSpanFactory	NoOpSpanFactory	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut84, Mark4TypeHDBSCAN1]	17
27	org.axonframework.eventhandling.EventBus	EventBus	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity10, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition10...	15
28	org.axonframework.messaging.annotation.HandlerDefinition	HandlerDefinition	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, ...	15
29	org.axonframework.common.transaction.NoTransactionManager	NoTransactionManager	[Type, File, Java, ByteCode, Enum, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity0, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut97, Mark4TypeLocalClusteringCoefficient0.14285714285714285, Mark4TypeHDBSCAN1]	15
30	org.axonframework.messaging.ResultMessage	ResultMessage	[Type, File, Java, ByteCode, Interface, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation18, Mark4TypeLouvainCommunit...	15
31	org.axonframework.tracing.Span	Span	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity6, ...	15
32	org.axonframework.messaging.unitofwork.DefaultUnitOfWork	DefaultUnitOfWork	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation18, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut98, Mark4TypeHDBSCAN117]	14
33	org.axonframework.common.ReflectionUtils	ReflectionUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation42, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition7,	14

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeMaximumKCut1, Mark4TypeHDBSCAN110, Mark4TopAnomalyHub]	
34	org.axonframework.messaging.InterceptorChain	InterceptorChain	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation55, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity11, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut23, Mark4TypeLocalClusteringCoefficient0.1038961038961039, Mark4TypeHDBSCAN-1]	13
35	org.axonframework.eventhandling.TrackedEventMessage	TrackedEventMessage	Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation31, Mark4TypeLouvainCommunity4, Mark4T...	13
36	org.axonframework.common.AxonException	AxonException	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation10, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity7, Mark4TypeKCoreDecomposit...	12
37	org.axonframework.common.DateTimeUtils	DateTimeUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN168]	12
38	org.axonframework.messaging.DefaultInterceptorChain	DefaultInterceptorChain	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity12, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut19, Mark4TypeLocalClusteringCoefficient0.25, Mark4TypeHDBSCAN168]	12
39	org.axonframework.eventhandling.GenericDomainEventMessage	GenericDomainEventMessage	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation18, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity12, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut0, Mark4TypeLocalClusteringCoefficient0.2727272727272727, Mark4TypeH...	12

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report `ArtifactPackageUsage` .

	artifactName	dependentArtifactName	dependentPackages	dependentArtifactPackages	packageUsagePercentage	dependentFullQualifiedPackageNames	dependentPackageNameNames
0	axon-tracing-opentelemetry-4.12.1	axon-messaging-4.12.1	3	70	0.042857	[org.axonframework.common, org.axonframework.messaging, org.axonframework.tracing]	[common, messaging, tracing]
1	axon-server-connector-4.12.1	axon-modelling-4.12.1	1	10	0.100000	[org.axonframework.modelling.command]	[command]
2	axon-spring-boot-autoconfigure-4.12.1	axon-test-4.12.1	1	8	0.125000	[org.axonframework.test.server]	[server]
3	axon-disruptor-4.12.1	axon-messaging-4.12.1	9	70	0.128571	[org.axonframework.commandhandling.callbacks, org.axonframework.messaging.unitofwork, org.axonframework.messaging.annotation, org.axonframework.common.transaction, org.axonframework.monitoring, org.axonframework.commandhandling, org.axonframework.common.caching, org.axonframework.messaging, org.---	[callbacks, unitofwork, annotation, transaction, monitoring, commandhandling, caching, messaging, common]
4	axon-test-4.12.1	axon-messaging-4.12.1	10	70	0.142857	[org.axonframework.eventhandling, org.axonframework.common, org.axonframework.messaging.annotation, org.axonframework.messaging, org.axonframework.messaging.unitofwork, org.axonframework.common.stream, org.axonframework.commandhandling, org.axonframework.deadline, org.axonframework.commandhandli...	[eventhandling, common, annotation, messaging, unitofwork, stream, commandhandling, deadline, gateway, scheduling]
5	axon-disruptor-4.12.1	axon-modelling-4.12.1	2	10	0.200000	[org.axonframework.modelling.command.inspection, org.axonframework.modelling.command]	[inspection, command]
6	axon-eventsourcing-4.12.1	axon-modelling-4.12.1	2	10	0.200000	[org.axonframework.modelling.command, org.axonframework.modelling.command.inspection]	[command, inspection]
7	axon-test-4.12.1	axon-eventsourcing-4.12.1	2	9	0.222222	[org.axonframework.eventsourcing, org.axonframework.eventstore, org.axonframework.eventsourcing]	[eventsourcing, eventstore]
8	axon-disruptor-4.12.1	axon-eventsourcing-4.12.1	2	9	0.222222	[org.axonframework.eventhandling, org.axonframework.messaging.annotation, org.axonframework.serialization.xml, org.axonframework.common.lock, org.axonframework.messaging.unitofwork, org.axonframework.common...	[eventhandling, annotation, xml, jdbc, serialization, common, lock, unitofwork, caching, tracing, legacyjpa, messaging, property, jpa, deadline, interceptors, commandhandling]
9	axon-modelling-4.12.1	axon-messaging-4.12.1	18	70	0.257143	[org.axonframework.messaging.unitofwork, org.axonframework.common, org.axonframework.commandhandling, org.axonframework.serialization, org.axonframework.common.io, org.axonframework.common.trans...	[unitofwork, annotation, common, commandhandling, eventhandling, messaging, serialization, io, transaction, event, jpa, jdbc, stream, lifecycle, xml, tracing, monitoring, lock, caching, legacyjpa]
10	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	20	70	0.285714	[org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing, org.axonframework.eventsourcing.snapshotting]	[eventstore, eventsourcing, snapshotting]
11	axon-server-connector-4.12.1	axon-eventsourcing-4.12.1	3	9	0.333333	[org.axonframework.messaging, org.axonframework.common, org.axonframework.serialization, org.axonframework.lifecycle, org.axonframework.commandhandling, org.axonframework.callbacks, org.axonframework.tracing, org.axonframework.eventh...	[messaging, common, serialization, lifecycle, commandhandling, callbacks, tracing, distributed, inmemory, queryhandling, eventhandling, transaction, tokenstore, responses, async, event, monitoring, java, xml, unitofwork, jdbc, scheduling, stream, util, pooled]
12	axon-server-connector-4.12.1	axon-messaging-4.12.1	25	70	0.357143	[org.axonframework.axonserver.connector, org.axonframework.axonserver.connector.command, org.axonframework.axonserver.connector.query, org.axonframework.axonserver.connector.event.axon]	[connector, command, query, axon]
13	axon-spring-boot-autoconfigure-4.12.1	axon-server-connector-4.12.1	4	11	0.363636	[org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.snapshotting, org.axonframework.eventsourcing, org.axonframework.eventsourcing.eventstore.jpa]	[eventstore, snapshotting, eventsourcing, jpa]
14	axon-configuration-4.12.1	axon-eventsourcing-4.12.1	4	9	0.444444	[org.axonframework.modelling.command.inspection, org.axonframework.modelling.command, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository.inmemory]	[inspection, command, repository, saga, inmemory]
15	axon-test-4.12.1	axon-modelling-4.12.1	5	10	0.500000	[org.axonframework.messaging.interceptors, org.axonframework.common.jpa, org.axonframework.common.lock, org.axonframework.common, org.axonframework.eventhandling, org.axonframework.eventhandling, org.axonframework.lifecycle, org.axonframework.serialization.upcasting.event, ...]	[interceptors, jpa, lock, common, deadletter, scheduling, lifecycle, event, tokenstore, correlation, deadline, unitofwork, xml, jdbc, gateway, inmemory, monitoring, serialization, annotation, queryhandling, tracing, util, eventhandling, async, caching, commandhandling, updates, messaging, pooled...
16	axon-configuration-4.12.1	axon-messaging-4.12.1	37	70	0.528571	[org.axonframework.eventsourcing.eventstore.legacyjpa, org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.eventstore.jpa, org.axonframework.eventsourcing, org.axonframework.eventsourcing.jdbc]	[legacyjpa, eventstore, jpa, eventsourcing, jdbc]
17	axon-spring-boot-autoconfigure-4.12.1	axon-eventsourcing-4.12.1	5	9	0.555556	[org.axonframework.modelling.saga.repository.legacyjpa, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga.repository.jdbc, org.axonframework.modelling.saga.repository.jpa, org.axonframework.modelling.saga.repository.command, org.axonframework.modelling.saga]	[legacyjpa, repository, jdbc, jpa, command, saga]
18	axon-spring-boot-autoconfigure-4.12.1	axon-modelling-4.12.1	6	10	0.600000	[org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga.repository.inmemory, org.axonframework.modelling.saga.repository.jpa]	[repository, saga, inspection, inmemory, command, jpa]
19	axon-configuration-4.12.1	axon-modelling-4.12.1	6	10	0.600000	[org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga.repository.inmemory, org.axonframework.modelling.saga.repository.command, org.axonframework.modelling.saga.repository.jpa]	[repository, saga, inspection, inmemory, command, jpa]
20	axon-modelling-4.12.1	axon-modelling-4.12.1	6	10	0.600000	[org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga.repository.inmemory, org.axonframework.modelling.saga.repository.command, org.axonframework.modelling.saga]	[saga, jpa, repository, metamodel, inspection, command]
21	axon-test-4.12.1	axon-test-4.12.1	5	8	0.625000	[org.axonframework.test.matchers, org.axonframework.test, org.axonframework.test.deadline, org.axonframework.test.eventscheduler, org.axonframework.test.utils]	[matchers, test, deadline, eventscheduler, utils]
22	axon-messaging-4.12.1	axon-messaging-4.12.1	44	70	0.628571	[org.axonframework.messaging.interceptors, org.axonframework.monitoring, org.axonframework.messaging.annotation, org.axonframework.common, org.axonframework.queryhandling, registration, org.axonframework.messaging.responses, org.axonframework.messaging, org.axonframework.tracing, org.axonfram...	[interceptors, monitoring, annotation, common, registration, responses, messaging, tracing, transaction, unitofwork, queryhandling, io, eventhandling, commandhandling, serialization, upcasting, stream, util, deadline, lifecycle, scheduling, jobrunr, digest, tokenstore, jdbc, xml, correlation...
23	axon-spring-boot-autoconfigure-4.12.1	axon-messaging-4.12.1	44	70	0.628571	[org.axonframework.messaging.timeout, org.axonframework.common.legacyjpa, org.axonframework.common.transaction, org.axonframework.eventhandling, org.axonframework.eventhandling, org.axonframework.messaging, org.axonframework.messaging, org.axonframework.serialization, org.axonframe...	[timeout, legacyjpa, transaction, tokenstore, deadletter, serialization, jdbc, eventhandling, jpa, scheduling, correlation, annotation, async, jobrunr, lifecycle, pooled, gateway, dscheduler, distributed, json, interceptors, common, avro, xml, event, commandhandling, messaging, attributes, upda...
24	axon-eventsourcing-4.12.1	axon-eventsourcing-4.12.1	7	9	0.777778	[org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.eventstore.jdbc, org.axonframework.eventsourcing.snapshotting, org.axonframework.eventsourcing, org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.jdbc, org.axonframework.eventsourcing.conflictresolution, org.axonfra...	[eventstore, jdbc, snapshotting, eventsourcing, statements, conflictresolution, jpa]

	artifactName	dependentArtifactName	dependentPackages	dependentArtifactPackages	packageUsagePercentage	dependentFullQualifiedPackageNames	dependentPackageNames
25	axon-server-connector-4.12.1	axon-server-connector-4.12.1	9	11	0.818182	[org.axonframework.axonserver.connector, org.axonframework.axonserver.connector.util, org.axonframework.axonserver.connector.command, org.axonframework.axonserver.connector.event.axon, org.axonframework.axonserver.connector.event.util, org.axonframework.axonserver.connector.query, org.axonframework...	[connector, util, command, axon, query, heartbeat, processor, subscription]
26	axon-spring-boot-autoconfigure-4.12.1	axon-spring-boot-autoconfigure-4.12.1	8	9	0.888889	[org.axonframework.springboot, org.axonframework.springboot.util, org.axonframework.springboot.util.legacyjpa, org.axonframework.springboot.autoconfigure, org.axonframework.actuator, org.axonframework.springboot.service.connection, org.axonframework.actuator.axonserver, org.axonframework.springboot...	[springboot, util, legacyjpa, autoconfig, actuator, connection, axonserver, jpa]
27	axon-spring-boot-autoconfigure-4.12.1	axon-configuration-4.12.1	1	1	1.000000	[org.axonframework.config]	[config]
28	axon-server-connector-4.12.1	axon-configuration-4.12.1	1	1	1.000000	[org.axonframework.config]	[config]
29	axon-configuration-4.12.1	axon-disruptor-4.12.1	1	1	1.000000	[org.axonframework.disruptor.commandhandling]	[commandhandling]

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts` .

	artifactName	dependentArtifactName	packageName	dependentPackage.fqn	dependentTypes	dependentPackageTypes	typeUsagePercentage
0	axon-spring-boot-autoconfigure-4.12.1	axon-messaging-4.12.1	org.axonframework.springboot.util	org.axonframework.eventhandling	1	100	0.010000
1	axon-spring-boot-autoconfigure-4.12.1	axon-messaging-4.12.1	org.axonframework.springboot.autoconfigure.legacyjpa	org.axonframework.eventhandling	1	100	0.010000
2	axon-modelling-4.12.1	axon-messaging-4.12.1	org.axonframework.modelling.saga.metamodel	org.axonframework.eventhandling	1	100	0.010000
3	axon-server-connector-4.12.1	axon-messaging-4.12.1	org.axonframework.axonserver.connector	org.axonframework.eventhandling	1	100	0.010000
4	axon-test-4.12.1	axon-messaging-4.12.1	org.axonframework.test.matchers	org.axonframework.eventhandling	1	100	0.010000
5	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.snapshotting	org.axonframework.eventhandling	1	100	0.010000
6	axon-eventsourcing-4.12.1	axon-modelling-4.12.1	org.axonframework.eventsourcing.conflictresolution	org.axonframework.modelling.command	1	56	0.017857 [org.axonframework.
7	axon-eventsourcing-4.12.1	axon-modelling-4.12.1	org.axonframework.eventsourcing.eventstore.jdbc	org.axonframework.modelling.command	1	56	0.017857 [org.i
8	axon-server-connector-4.12.1	axon-modelling-4.12.1	org.axonframework.axonserver.connector	org.axonframework.modelling.command	1	56	0.017857 [org.i
9	axon-test-4.12.1	axon-messaging-4.12.1	org.axonframework.test.eventscheduler	org.axonframework.eventhandling	2	100	0.020000
10	axon-modelling-4.12.1	axon-messaging-4.12.1	org.axonframework.modelling.command	org.axonframework.eventhandling	2	100	0.020000 [org.a
11	axon-modelling-4.12.1	axon-messaging-4.12.1	org.axonframework.modelling.command.legacyjpa	org.axonframework.eventhandling	2	100	0.020000 [org.a
12	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.conflictresolution	org.axonframework.eventhandling	2	100	0.020000
13	axon-spring-boot-autoconfigure-4.12.1	axon-configuration-4.12.1	org.axonframework.springboot	org.axonframework.config	1	43	0.023256
14	axon-spring-boot-autoconfigure-4.12.1	axon-configuration-4.12.1	org.axonframework.springboot.autoconfigure.legacyjpa	org.axonframework.config	1	43	0.023256
15	axon-server-connector-4.12.1	axon-configuration-4.12.1	org.axonframework.axonserver.connector.processor	org.axonframework.config	1	43	0.023256
16	axon-server-connector-4.12.1	axon-eventsourcing-4.12.1	org.axonframework.axonserver.connector.event.axon	org.axonframework.eventsourcing	1	42	0.023810
17	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.conflictresolution	org.axonframework.messaging	1	35	0.028571
18	axon-test-4.12.1	axon-messaging-4.12.1	org.axonframework.test.matchers	org.axonframework.messaging	1	35	0.028571
19	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.eventstore	org.axonframework.messaging	1	35	0.028571
20	axon-test-4.12.1	axon-messaging-4.12.1	org.axonframework.test	org.axonframework.messaging	1	35	0.028571
21	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.eventstore.legacyjpa	org.axonframework.serialization	1	34	0.029412
22	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.eventstore.jdbc	org.axonframework.serialization	1	34	0.029412
23	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.eventstore.jpa	org.axonframework.serialization	1	34	0.029412
24	axon-spring-boot-autoconfigure-4.12.1	axon-messaging-4.12.1	org.axonframework.springboot.autoconfigure.legacyjpa	org.axonframework.serialization	1	34	0.029412
25	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.eventstore.jdbc.statements	org.axonframework.eventhandling	3	100	0.030000 [org.a
26	axon-test-4.12.1	axon-modelling-4.12.1	org.axonframework.test.utils	org.axonframework.modelling.saga	1	33	0.030303 [
27	axon-eventsourcing-4.12.1	axon-messaging-4.12.1	org.axonframework.eventsourcing.conflictresolution	org.axonframework.commandhandling	1	32	0.031250
28	axon-spring-boot-autoconfigure-4.12.1	axon-server-connector-4.12.1	org.axonframework.springboot.service.connection	org.axonframework.axonserver.connector	1	32	0.031250 [org.axon
29	axon-spring-boot-autoconfigure-4.12.1	axon-server-connector-4.12.1	org.axonframework.actuator.axonserver	org.axonframework.axonserver.connector	1	32	0.031250 [org.axonframe

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts` .

packageName	duplicates	artifactNames
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Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

[illegible]

Table 10 - Distance distribution between dependent files

This table shows the file directory distance distribution between dependent files. Intuitively, the distance is given by the fewest number of change directory commands needed to navigate between a file and a dependency it uses. Those are aggregate to see how many dependent files are in the same directory, how many are just one change directory command apart, and so on.

dependency.fileDistanceAsFewestChangeDirectoryCommands	numberOfDependencies	numberOfDependencyUsers	numberOfDependencyProviders	examples
0	0	2719	1080	1216
1	1	95	85	41
2	2	2583	786	487
3	4	2655	460	562